

Md. Almabud **JUWAL**

Senior Software Engineer | B.Sc. Computer Science and Engineering

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 Dhaka, Bangladesh

 EDUCATION

|      |                                                                                                                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 | B.Sc. in Computer Science and Engineering                                                                                                                                                                                                                                                                                                  |
| 2016 | <div><div>&gt; Daffodil International University, Dhaka, Bangladesh</div><div>&gt; CGPA : 3.77 / 4.00</div><div>&gt; Relevant coursework : Artificial Intelligence, Algorithms, Data Structures, Software Engineering, Compiler Design, Discrete Mathematics, Computer Networks</div><div>&gt; Medium of instruction : English</div></div> |

 ACADEMIC & RESEARCH INTERESTS

INTERESTS :   Artificial Intelligence, Machine Learning, Data Science,  
Software Engineering, Distributed Systems

 SKILLS

|                              |                                                          |
|------------------------------|----------------------------------------------------------|
| Programming                  | Python, SQL, JavaScript/TypeScript, Go, C/C++            |
| Backend & Frameworks         | FastAPI, Django, Django REST Framework, Celery           |
| Data & Databases             | PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch |
| Infrastructure & Engineering | Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible           |
| Frontend                     | Angular                                                  |

 EXPERIENCE

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| Present<br>Jun 2024  | <div>Senior Software Engineer   Cogwheel Analytics, Colorado, United States</div> <div><div>&gt; Architected and led the development of the next-generation <b>Cogwheel Analytics</b> platform as a <b>modular monolith</b> following <b>Clean Architecture</b> principles, with clear separation of concerns and reusable handling for authentication, pagination, and request validation, reducing repetitive implementation work and accelerating feature delivery</div><div>&gt; Lead the development and maintenance of the <b>Data Cleaning</b> service, implementing data-validation and transformation workflows to improve data integrity and consistency</div><div>&gt; Design and implement backend services and APIs using Python, FastAPI, and PostgreSQL, with emphasis on maintainability, reliability, and performance</div><div>&gt; Build and maintain asynchronous data-processing and web-scraping workflows using Pandas, Celery, and Scrapy</div></div> <div><div>Python</div><div>FastAPI</div><div>PostgreSQL</div><div>Pandas</div><div>Celery</div><div>Scrapy</div><div>Angular</div></div> |
| May 2024<br>Jul 2023 | <div>Senior Software Engineer   Brainstation 23 Ltd, Dhaka, Bangladesh</div> <div><div>&gt; Led the turnaround of a delayed project by mentoring <b>3 junior engineers</b>, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within <b>6 months</b></div><div>&gt; Designed software architectures for <b>Task Station</b> and <b>Sayvery</b>, with focus on scalability, maintainability, and clear separation of responsibilities</div><div>&gt; Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines</div><div>&gt; Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction</div></div> <div><div>Python</div><div>Django</div><div>Django REST Framework</div><div>PostgreSQL</div><div>Celery</div><div>Scrapy</div></div>                                                                                                                                     |

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| Jun 2023<br>Oct 2021 | <b>Software Engineer   Brainstation 23 Ltd, Dhaka, Bangladesh</b> <ul style="list-style-type: none"> <li>Improved system performance by approximately <b>10%</b> through database-query optimization</li> <li>Improved API response performance by approximately <b>5%</b> by introducing application-level caching</li> <li>Upgraded Python and Django versions and refactored the existing codebase to improve maintainability and compatibility</li> <li>Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer</li> </ul> <div>Python Django Django REST Framework PostgreSQL Redis Celery Go</div>                                                                             |
| Sep 2021<br>Jul 2020 | <b>Junior Software Engineer   Tirezok Private Limited, Dhaka, Bangladesh</b> <ul style="list-style-type: none"> <li>Reduced processing time for <b>3.3 million records</b> from approximately <b>29–35 minutes to 2–4 minutes</b> using PostgreSQL COPY and database triggers</li> <li>Reduced database query response time by approximately <b>50%</b> through query optimization</li> <li>Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j</li> <li>Automated deployment workflows using Jenkins and Ansible, reducing manual deployment effort and improving release consistency</li> </ul> <div>Python Django PostgreSQL Redis Neo4j Jenkins Ansible Angular</div> |
| Apr 2020<br>Dec 2019 | <b>Web Developer   Wizdoor Inc, Dhaka, Bangladesh</b> <ul style="list-style-type: none"> <li>Contributed to backend and database architecture while coordinating implementation with frontend and design teams</li> <li>Developed a real-time chat application using Django Channels and WebSockets</li> </ul> <div>Python Django Django REST Framework PostgreSQL Django Channels Angular</div>                                                                                                                                                                                                                                                                                                                                                                                   |

## PROJECTS

### COGWHEEL ANALYTICS

Jun 2024 – Present

<https://cogwheelanalytics.com>

Enterprise analytics platform for automated data ingestion, validation, processing, and reporting. The next-generation platform is designed as a modular monolith following Clean Architecture principles, with reusable application components and asynchronous processing to support maintainable and scalable analytics workflows.

Python FastAPI PostgreSQL Pandas Celery RabbitMQ Scrapy Angular

### REHIVE

Oct 2021 – Feb 2023

<https://rehive.com/>

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

Python Django Django REST Framework PostgreSQL Celery RabbitMQ Redis Docker Angular

### LEGISLIB

May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

Python Django Django REST Framework MongoDB Elasticsearch PDF Processing

## ACADEMIC & PROGRAMMING ACHIEVEMENTS

|      |                                                               |
|------|---------------------------------------------------------------|
| 2018 | Participant, ACM ICPC Dhaka Preliminary Contest               |
| 2018 | <b>4th place</b> , DIU Intra Team Programming Contest         |
| 2017 | <b>2nd place</b> , DIU Intra Team Programming Contest         |
| 2017 | <b>9th place</b> , DIU Takeoff Individual Programming Contest |

## CERTIFICATIONS & COURSES

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Jun 2025 | Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka |
| Apr 2023 | AWS Certified Cloud Practitioner                                                  |
| 2021     | Object-Oriented Design, Coursera                                                  |
| 2021     | The Ultimate Docker Course, Mosh Hamedani                                         |

## TEACHING AND MENTORING

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| Dec 2018 | ACM Co-Chair   Computer & Programming Club, Daffodil International University                                                                                                                                                                                                                                                                                |
| Jan 2018 |                                                                                                                                                                                                                                                                                                                                                              |
|          | <ul style="list-style-type: none"> <li>&gt; Coordinated programming contests and other ACM-related club activities</li> <li>&gt; Mentored students in competitive programming and encouraged participation in programming contests</li> </ul>                                                                                                                |
| Jan 2019 | Student Prefect   Daffodil International University                                                                                                                                                                                                                                                                                                          |
| May 2017 |                                                                                                                                                                                                                                                                                                                                                              |
|          | <ul style="list-style-type: none"> <li>&gt; Assisted students with Competitive Programming and Algorithms through practical lab sessions and academic support</li> <li>&gt; Assisted the course instructor with laboratory examinations and student performance evaluation</li> <li>&gt; Mentored students requiring additional academic guidance</li> </ul> |

## LANGUAGES

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- > **Bengali** — Native
- > **English** — B2 (CEFR), IELTS Academic 6.5
- > **Hindi** — Conversational